This Candidate is [MASK] Prompt-based Sentiment Extraction and Reference Letters

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Sentiment Analysis Using PLMs

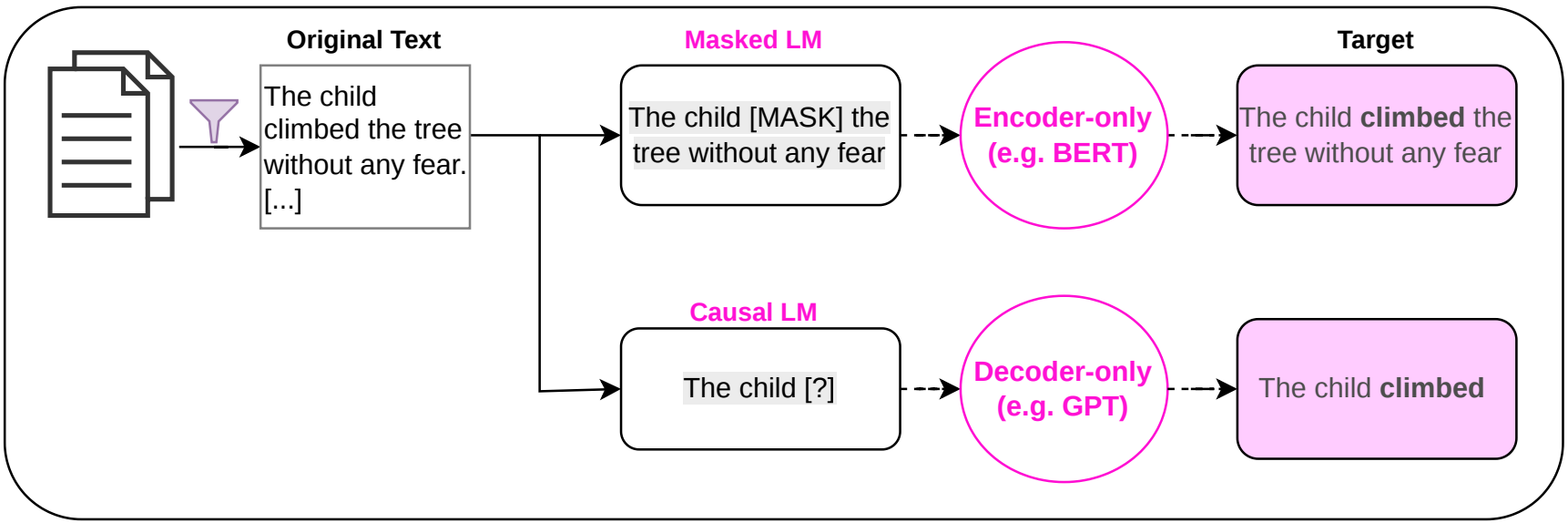
- I propose a relatively simple way to deploy pre-trained language models (PLMs) in order to extract sentiment and other useful features from text data
- I refer to this method as *prompt-based sentiment extraction*
- A textual prompt can be devised to elicit a sentiment prediction without the need to fine-tune the model

Example

To recognize the emotion of the tweet "I missed the bus today!" we may continue with the prompt "I felt so [MASK]", and ask the PLM to fill the blank with an emotion-bearing word.

Pre-training

- Self-supervised learning
  - Unlabeled data (e.g. text from the internet)
  - Biased, incorrect and misspelled text $\Rightarrow$ **filter**
- Tasks custom-designed for the model architecture



Advantages of Prompt-based Sentiment Extraction

- Unlike bag-of-words approaches:
  - Natural language(s) accepted "as is" — no need for heavy-handed pre-processing of the text input
  - Words receive a representation that is highly *context-dependent*
  - No need to compose a *lexicon* and no ad-hoc aggregation of scores
- Unlike fine-tuned LMs:
  - No need for labeled data or fine-tuning
  - No domain specificity
- Unlike instruct models/chatbots:
  - Sentiment score gets a probability interpretation
  - Reproducible results
- Other
  - Prompting can be tested using text meta-data
  - Features of text other than sentiment may be extracted as well

An Application to Academic Reference Letters

- I apply prompt-based sentiment extraction to a hand-collected corpus of reference letters (RLs)
- I validate the method by predicting the candidate's sex and field of research
- I show that the *average* sentiment of the candidate's RLs positively correlates with job market outcomes
- Sentiment *dispersion*—i.e. disagreement among letter writers—has negative impact
- I compare the performance of other methods (lexical approaches, supervised sentiment models, and chatbots) in predicting success. No other sentiment extraction method could produce similar results
- I show that RLs written for female candidates use more *grindstone* and less *standout* language

Prompt-based Learning

Three steps:

1. Apply a *prompting function* $f_{prompt}(\cdot)$ to modify the input text $\boldsymbol{x}$ into a *prompt*: $\boldsymbol{x}' = f_{prompt}(\boldsymbol{x})$
   - A template is a text string that has two slots: an input slot [X] for input $\boldsymbol{x}$ and an answer slot [MASK] for an intermediate *answer text* $\boldsymbol{m}$.

Example

Template: "[X] Overall, it was a [MASK] movie."

$\boldsymbol{x}'$ = "I love this movie. Overall, it was a [MASK] movie."

Three steps:

2. Define a *verbalizer* ($\mathcal{V}(\boldsymbol{m})$).
   - Let $\mathcal{M}$ be the set of permissible answers.
   - Example:

$$\mathcal{M} = \{\text{“excellent”}; \text{“good”}; \text{“OK”}; \text{“bad”}; \text{“horrible”}\}$$

   .
   - A verbalizer is a mapping ($\mathcal{V}(\boldsymbol{m}) : \mathcal{M} \to \mathcal{Y}$) assigning each permissible answer to a label.

Example

$$\begin{aligned}\mathcal{V} = \{&(\text{“excellent”}, +); (\text{“good”}, +); (\text{“OK”}, \sim);\\ &(\text{“bad”}, -); (\text{“horrible”}, -)\}\end{aligned}$$

Three steps:

3. Collect predicted probabilities for each label.
   - Let $f_{fill}(\boldsymbol{x}', \boldsymbol{m})$ fill-in the location [MASK] in prompt $\boldsymbol{x}'$ with the potential answer $\boldsymbol{m}$.

$$\mathrm{P}(y \mid \boldsymbol{x}) = \sum_{\mathbf{m} \in \mathcal{M}} \mathrm{I}\left[\mathcal{V}(\boldsymbol{m}) = y\right] \mathrm{P}\left[f_{fill}(f_{prompt}(\boldsymbol{x}), \boldsymbol{m}); \boldsymbol{\gamma}\right], \quad y \in \mathcal{Y}$$

   - Probabilities can be used to predict a label or to compute other statistics

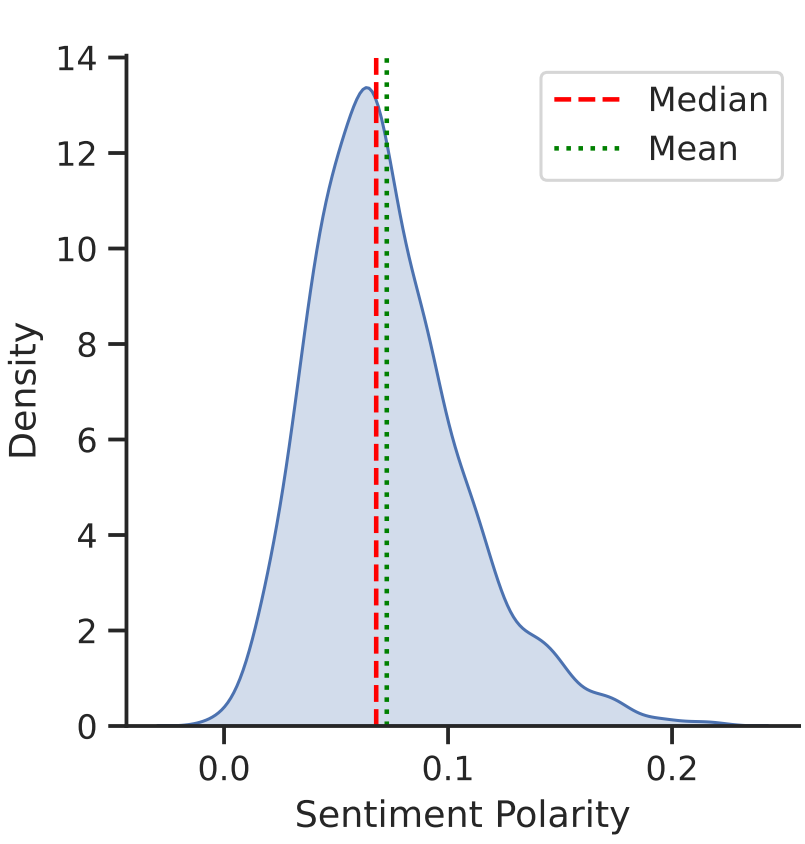
Example

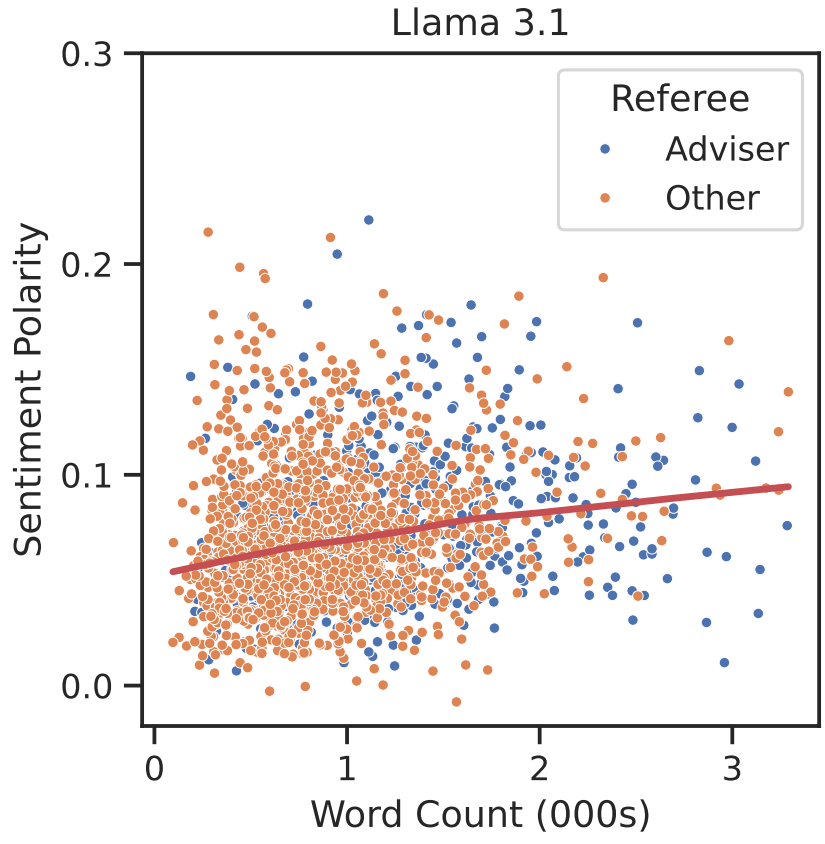
$$\mathit{Polarity}(\boldsymbol{x}) = \mathrm{P}(y = + \mid \boldsymbol{x}) - \mathrm{P}(y = - \mid \boldsymbol{x})$$

Prompt-based Sentiment Extraction from RLs

Prompt ($\mathcal{T}_{sentiment}$)	**Verbalizer** ($\mathcal{V}_{sentiment}$)	
"[X] In summary, this job market candidate is [MASK]"	**positive**:	{excellent, outstanding, exceptional, brilliant, great, superb, fantastic, terrific, amazing, phenomenal, awesome, stellar, wonderful, extraordinary, remarkable, impressive, incredible, fabulous, formidable, marvelous, perfect, fascinating}
	negative:	{bad, terrible, disappointing, poor, awful, lousy, atrocious, pathetic, unsatisfactory, substandard, mediocre, common, inferior, average, ordinary, regular}

The Distribution of Prompt-based RL Sentiment

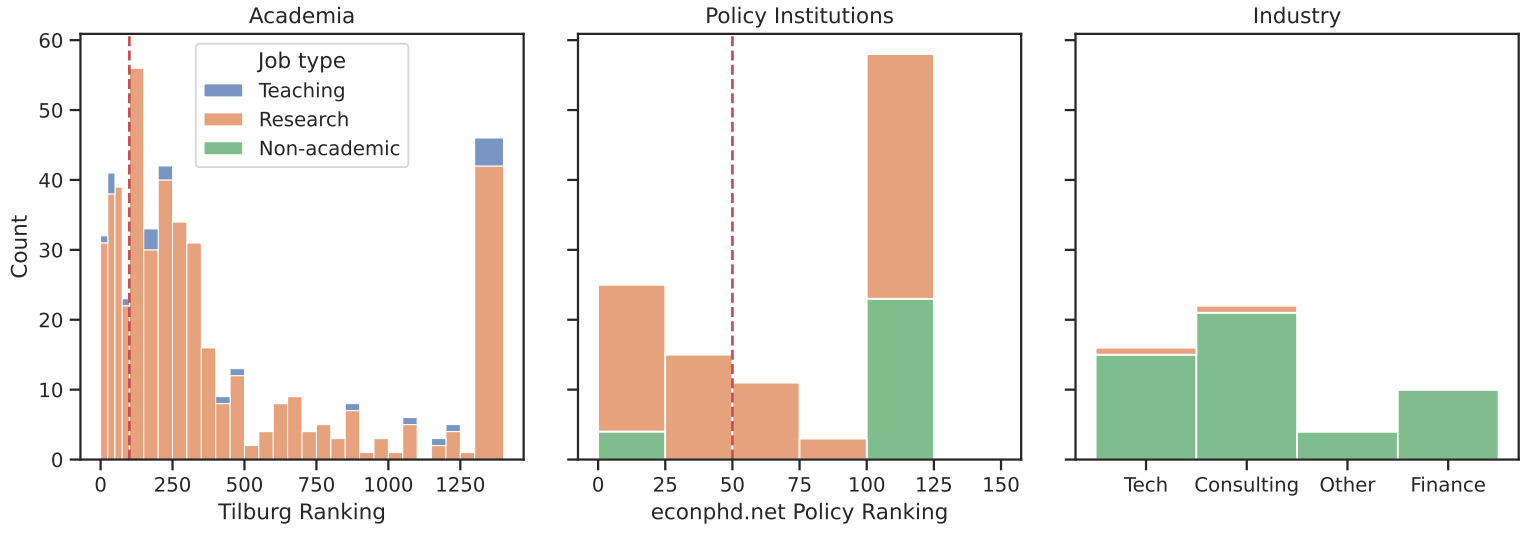




Notes: N=1968. Fitted line is a locally weighted regression.

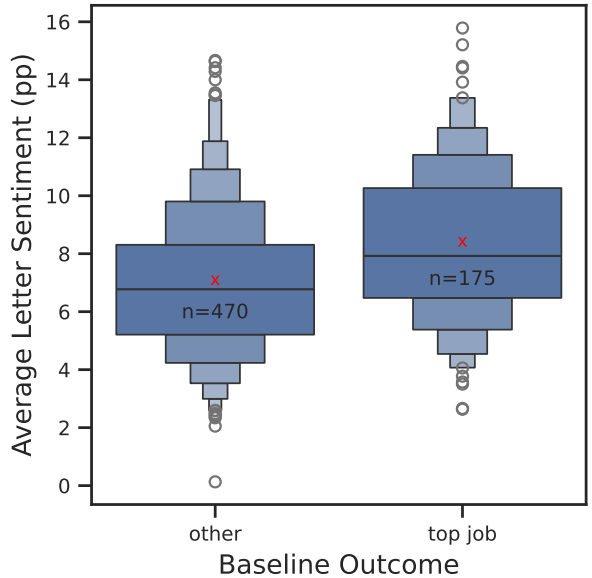
Predicting Job Market Outcomes

- Jobs vary by employer type, position, and institution ranking:



Outcome	**Definition**	$\%_{success}$
Baseline	Top-100 Academic Employers & Top-50 Policy Institutions	27.1
Alt. 1	Exclude teaching positions (liberal arts colleges) and post-docs from the success group	17.8
Alt. 2	Add some highly paid tech jobs (Amazon, Meta, etc.) to the success group	29.5
Alt. 3	Use somewhat more forgiving cutoffs: top-150 academic and top-75 policy	37.7
Alt. 4	Use somewhat more restrictive cutoffs: top-75 academic and top-35 policy	23.3

Llama Predicts Job Market Outcomes



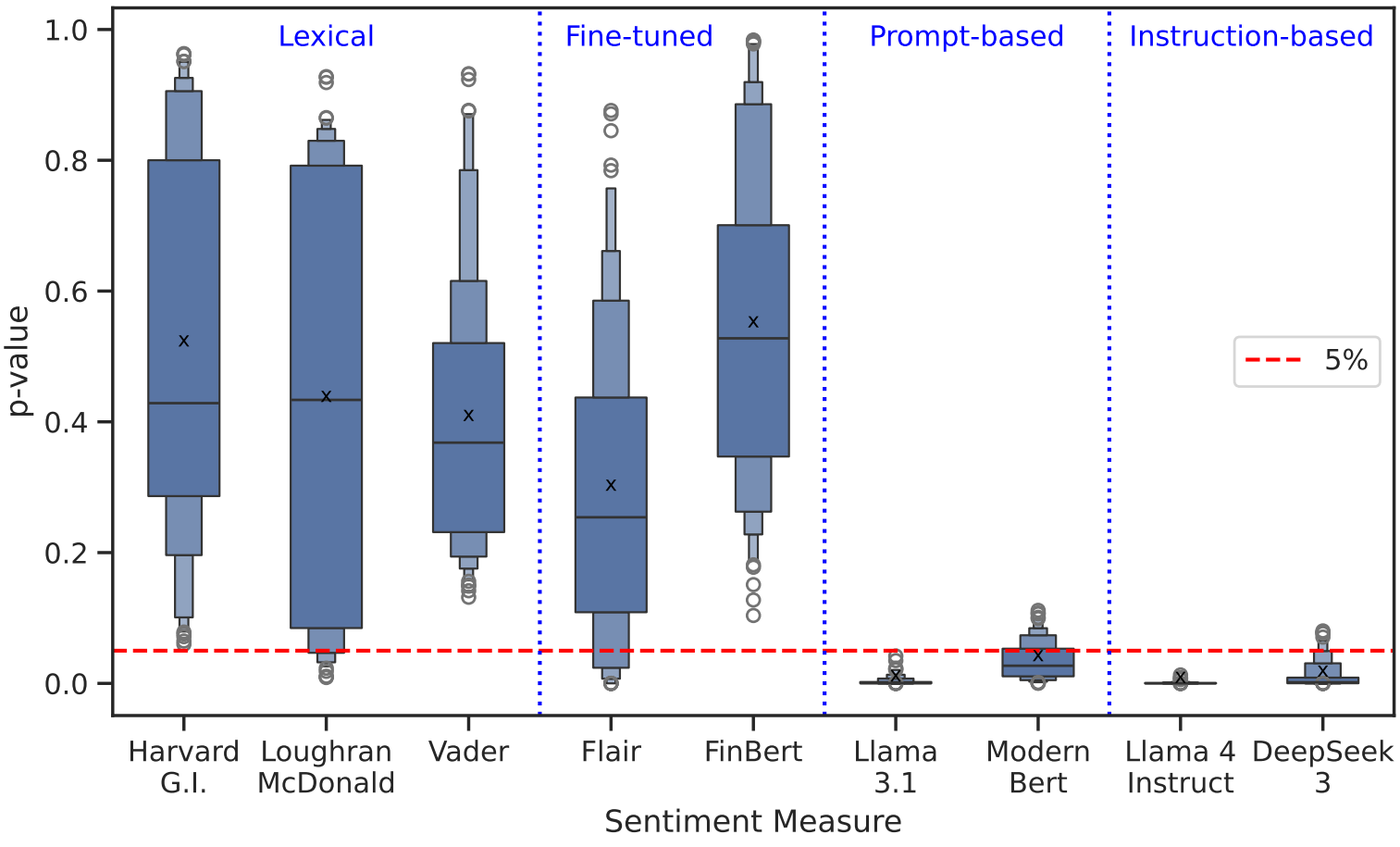
Outcomes $\rightarrow$	*Baseline*	*Alt. 1*	*Alt. 2*	*Alt. 3*	*Alt. 4*
Avg. RL Sentiment (p.p.)	0.0227 (0.0074)*** [0.0064]*** |0.0098|**	0.0147 (0.0064)** [0.0057]*** |0.0072|**	0.0237 (0.0077)*** [0.0068]*** |0.0088|***	0.0276 (0.0086)*** [0.0081]*** |0.0098|***	0.0229 (0.0073)*** [0.0064]*** |0.0093|**
RL Count & Avg. Length	Yes	Yes	Yes	Yes	Yes
Candidate's characteristics	Yes	Yes	Yes	Yes	Yes
PhD characteristics	Yes	Yes	Yes	Yes	Yes
Job Market Year	Yes	Yes	Yes	Yes	Yes
Number of top-5% Writers	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.13	0.13	0.12	0.14	0.11

Notes: $N = 645$. *p<0.10; **p<0.05; ***p<0.01.
Standard Errors: (HC3) [Clustered: Candidate Univ.] |Clustered: Univ Rank×Period|.

Alternative Sentiment Measures

p-value

- For each sentiment measure, we have 120 regressions: 5 outcomes, 3 standard errors, 2 samples, 4 control variable groups



Sentiment Dispersion and Job Market Outcomes

	Baseline Outcome		
	Range	SD	MAD
Average Letter Sentiment (percent)	0.0368***	0.0366***	0.0360***
	(0.0087)	(0.0093)	(0.0092)
	[0.0080]	[0.0080]	[0.0079]
	|0.0107|	|0.0108|	|0.0108|
Sentiment Range (max-min)	-0.0219***		
	(0.0077)		
	[0.0077]		
	|0.0073|		
Sentiment SD		-0.0419***	
		(0.0153)	
		[0.0151]	
		|0.0147|	
Sentiment MAD			-0.0532***
			(0.0204)
			[0.0201]
			|0.0200|
Four Letters – Avg. Letter Length	Yes	Yes	Yes
Candidate's characteristics	Yes	Yes	Yes
PhD characteristics	Yes	Yes	Yes
Job Market Year	Yes	Yes	Yes
Number of top-5% Writers	Yes	Yes	Yes
Adjusted R^2	0.132	0.131	0.130

Notes: $N = 553$. ***p<0.01.
Standard Errors: (HC3) [Cluster: Candidate Univ.] |Cluster: Univ Rank×Period|.

Prompt-based Gendered Sentiment Extraction

Prompt ($\mathcal{T}_{personality}$)	**Verbalizer** ($\mathcal{V}_{personality}$)	
"[X] If I had to pick this candidate's main strength, I would mention their [MASK]"	**standout**:	{ability, research, creativity, originality, leadership, skill, skills, intelligence, theoretical, technical, mathematical, analytical, statistical, intellectual, original, critical, ambition, drive, excellence}
	grindstone:	{passion, dedication, diligence, patience, perseverance, persistence, open, personality, genuine, interpersonal, collaborative, enthusiasm, work, attention, humility, communication, maturity, flexibility, commitment, writing, teaching}

Gendered Sentiment Scores

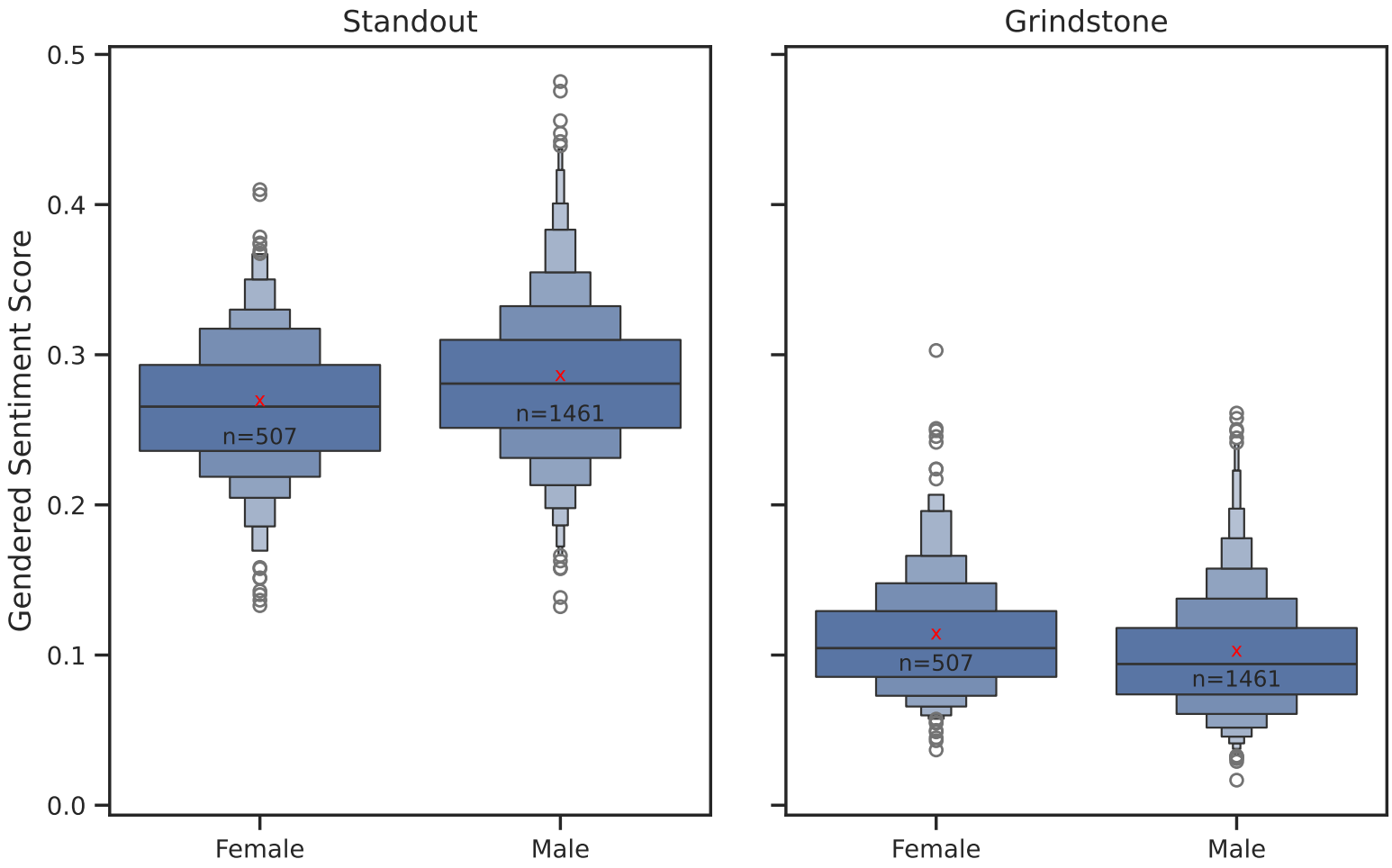


Figure: Gender Differences in Sentiment

Conclusions

- Prompt-based learning is a promising new way to extract information from text data
  - PLMs take context into account
  - The text pre-processing necessary is modest compared to previous methods
  - Prompt-based sentiment scores have a probability interpretation
  - Model performance can be benchmarked against known meta-data
  - Fully reproducible results
  - The power of transformers is only going to increase going forward
  - Challenges: prompt and verbalizer engineering

Conclusions

- There is compelling evidence that reference letter sentiment is predictive of job market outcomes
  - Average sentiment polarity is positively associated with the probability of success in the job market
  - The dispersion of letter writer sentiment negatively affects outcomes
- There are small but consequential biases in how male and female candidates are described in RLs

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